

The row with two $\mathfrak{C}$ vertices

Assembly check — where the transverse divergence really is — dimensional regularisation of the z integral — the p^+ integral with a $+$ prescription

$$S \equiv p^+ + k^+, \quad r \equiv y' - y, \quad \kappa \equiv \frac{k \cdot r}{k^+}, \quad P_V \equiv V - k^+, \quad d = 2 - 2\epsilon$$

Verdict in one paragraph. The prefactor, the two $\delta^{(2)}$'s, the w, w' integrations and the full tensor contraction of the two $\mathfrak{C}$ brackets are all **correct** — every step reproduces your quoted four-term bracket exactly (§1). Four bookkeeping corrections are needed: a missing N_c from the colour reduction, two transposed adjoint indices, $\int_{z,w,w'} \rightarrow \int_z$ in the last two lines, and a missing x' in the measure. On the divergences: **there is no ultraviolet divergence anywhere in the transverse integrand — and in particular none at $y \rightarrow y'$** (§2). What is divergent is the **infrared** end of the z integral, $|z| \rightarrow \infty$, because the colour reduction has removed every $U(z)$ and the phase never touches z . Dimensional regularisation in $d = 2 - 2\epsilon$ with $\epsilon > 0$ handles it, and — this is the point — the same $\frac{1}{\epsilon}(r^2)^{-\epsilon}$ that carries the pole is the $\log \frac{1}{(y-y')^2}$ you were worried about (§4). The $2/k^+$ term then cancels identically (§5), and the $+$ prescription goes through cleanly on what is left (§6). Final result boxed in §7.

1. The assembly is correct

1.1 Prefactor

With $\mathfrak{A}^\dagger = -\frac{ig}{\sqrt{2}\pi\sqrt{S}}$, $\mathfrak{A} = +\frac{ig}{\sqrt{2}\pi\sqrt{S}}$ and two $\mathfrak{C}$'s each carrying $-\frac{g}{2\pi\sqrt{2S}} \frac{\sqrt{p^+(S-p^+)}}{S}$:

$$(-i)(-1)(-1)(+i) = +1 \quad (\text{real, as your } +\frac{g^4}{4\pi^4} \text{ requires}),$$

$$\frac{4}{(2\pi)^3} \cdot \frac{1}{(\sqrt{2}\pi)^2} \cdot \frac{1}{(2\pi\sqrt{2})^2} = \frac{1}{32\pi^7} = \frac{1}{(2\pi)^3} \cdot \frac{1}{4\pi^4} \checkmark,$$

$$\frac{1}{S} \cdot \frac{p^+(S-p^+)}{S^3} = \frac{p^+k^+}{(p^+ + k^+)^4} \checkmark \quad (S - p^+ = k^+).$$

1.2 The two deltas and the w, w' integrations

$$\delta^{(2)}\left[y - w + \frac{p^+}{S}(w - z)\right] \implies w = \frac{S y - p^+ z}{k^+}, \quad w - z = \frac{S}{k^+}(y - z),$$

so each WW kernel $\frac{(w-z)^m}{(w-z)^2} \rightarrow \frac{k^+}{S} \frac{(y-z)^m}{(y-z)^2}$ (two of them) and each delta contributes a Jacobian $(S/k^+)^2$ (two of them):

$$\frac{p^+k^+}{S^4} \cdot \left(\frac{k^+}{S}\right)^2 \cdot \left(\frac{S}{k^+}\right)^4 = \frac{p^+}{k^+S^2} \checkmark$$

which is exactly your overall $\frac{1}{k^+}$ with a $\frac{p^+}{S^2}$ left inside the bracket. And $w' - w = \frac{S}{k^+}(y' - y)$ gives the phase $e^{-ik \cdot (y' - y)S/k^+} = e^{-ik \cdot (y' - y)} e^{-ik \cdot (y' - y)p^+/k^+}$ — your third line. $\checkmark$

1.3 The tensor contraction

Relabelling the slots of $\mathfrak{C}_{1jik}$ gives precisely your bracket $[\delta_{km}\delta_{ij} - \frac{S}{p^+}\delta_{jm}\delta_{ik} - \frac{S}{k^+}\delta_{im}\delta_{kj}]$. Contracting the bra and ket brackets over the internal i, j :

$$T^{mm'kk'} = \left[d_\perp - \frac{2S^2}{p^+k^+}\right]\delta_{km}\delta_{k'm'} + \left[\frac{S^2}{p^{+2}} + \frac{S^2}{k^{+2}}\right]\delta_{mm'}\delta_{kk'} + \frac{2S^2}{p^+k^+}\delta_{km'}\delta_{k'm},$$

and multiplying by the $\frac{p^+}{S^2}$ of §1.2 reproduces your four terms **exactly** (symbolic difference = 0 for all three structures):

$$\frac{d_\perp p^+}{S^2}\delta_{km}\delta_{k'm'} + \frac{2}{k^+}(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \left(\frac{p^+}{k^{+2}} + \frac{1}{p^+}\right)\delta_{mm'}\delta_{kk'}.$$

1.4 Four corrections

1. **A factor N_c is missing.** The reduction uses $U^{db}(z)U^{db}(z) = \delta^{dd}$ and then $f^{cda}f^{cda} = N_c\delta^{cc}$. Checked numerically with explicit $SU(3)$ f^{abc} and random adjoint Wilson lines: the full colour contraction equals **exactly** $3 \times$ your reduced form (max deviation 2.7×10^{-15}), and $3 = N_c$.
2. **Two transposed indices.** As literally written, the ket has $U^{db}(z)$ in the first term but $U^{bd}(z)$ in the second, so the cross terms give $(U(z)U(z))^{dd}$, not δ^{dd} (numerically off by 1.67). The same mismatch appears as $U^{ce}(x)$ against $U^{ec}(y)$. With the orderings made uniform the reduction is exact. So the last line should read $[U^{ec}(y') - U^{ec}(x')][U^{ec}(y) - U^{ec}(x)]$, not $\dots - U^{ce}(x)$.
3. $\int_{z,w,w'} \rightarrow \int_z$ in the last two displayed equations — w and w' have already been eaten by the deltas.
4. $\int_{x,y,y'} \rightarrow \int_{x,x',y,y'}$ — x' appears in the integrand.

2. Where the transverse singularities are — and there is no UV

Power counting: if n points collapse onto each other at scale ρ the measure supplies $\rho^{2(n-1)} \frac{d\rho}{\rho}$, against an integrand $\rho^{-\nu}$ with ν the number of kernels that go singular. A logarithm needs the product to be ρ^0 . The four kernels here live on the separations $\{y, z\}$, $\{x, y\}$, $\{y', z\}$, $\{x', y'\}$ — **no kernel connects y to y'** , and the phase $e^{-ik \cdot (y' - y)S/k^+}$ is analytic there. Direct scan of the angular-averaged integrand times the measure:

collapsing region	$\rho = 10^{-2}$	10^{-4}	10^{-6}	scaling	verdict
$y' \rightarrow y$ (your suspicion)	6.29e-5	6.29e-9	6.29e-13	ρ^2	convergent
$z \rightarrow y$	1.210e-2	1.210e-4	1.210e-6	ρ^1	convergent
$z \rightarrow y'$	1.198e-2	1.198e-4	1.198e-6	ρ^1	convergent
$x \rightarrow y$	1.175e-2	1.175e-4	1.175e-6	ρ^1	convergent
$x' \rightarrow y'$	5.70e-3	5.70e-5	5.70e-7	ρ^1	convergent
$y', z \rightarrow y$ (3 pts)	6.17e-5	6.17e-9	6.17e-13	ρ^2	convergent
$x, y', z \rightarrow y$ (4 pts)	6.11e-7	6.11e-13	6.11e-19	ρ^6	convergent
$x, x', y', z \rightarrow y$ (5 pts)	8.48e-9	8.48e-17	8.48e-25	ρ^8	convergent

A constant column would mean a logarithm; a growing one a power divergence. Every column falls. **Not a single UV divergence, and $y \rightarrow y'$ is the most convergent region of all (ρ^2)**: there is simply nothing singular connecting y and y' in the integrand.

The divergence is at the other end. Send $|z| \rightarrow \infty$ with everything else fixed. Then $\frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2} \rightarrow \frac{\hat{z}^m \hat{z}^{m'}}{|z|^2}$, the measure is $|z|^2 \frac{d|z|}{|z|}$, the phase never involves z , and after the colour reduction **no $U(z)$ survives to cut the integral off**. Numerically the angular average times $|z|^2$ flatlines:

$ z $	10	10^2	10^3	10^4	10^5
$\langle \text{integrand} \rangle z ^2$	-0.1863	-0.18596	-0.185959	-0.185959	-0.185959
	+2.9898i	+2.98380i	+2.983732i	+2.983732i	+2.983732i

A perfect constant to six digits $\Rightarrow$ a logarithmic **infrared** divergence.

3. The z integral in closed form (cutoff)

All of the z dependence sits in the rank-2 tensor

$$T^{mm'}(y, y') = \int d^2 z \frac{(y-z)^m}{(y-z)^2} \frac{(y'-z)^{m'}}{(y'-z)^2}.$$

Using the complex-coordinate masters $\int_{|z|<R} \frac{d^2 z}{(\bar{z}-\bar{a})(z-b)} = \pi \log \frac{R^2}{|a-b|^2}$ for the trace and $\int_{|z|<R} \frac{d^2 z}{z-\bar{y}} = -\pi \bar{y}$ for the traceless part:

$$T^{mm'} = \frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'}, \quad r = y' - y.$$

Verified by 2D quadrature with $R = 3000$ at a generic non-aligned configuration:

```
numeric T = [[24.05094, 1.54771], [1.54036, 23.47890]]
closed T = [[24.03486, 1.54505], [1.54505, 23.46834]]
antisymmetric part of the numeric T : 7.3e-03 -> T is SYMMETRIC
```

Two features to read off. First, the $\log R^2$ is the IR divergence of §2. Second, $-\log r^2$: **this is the $y \rightarrow y'$ singularity you sensed**. But it is a *logarithm of the result*, not a divergence of an integral — and $\int d^2 y' \log \frac{1}{(y-y')^2}$ is perfectly convergent. It is the same object as the IR pole, as the next section makes manifest.

4. Dimensional regularisation, $d = 2 - 2\epsilon$

Take $\epsilon > 0$, i.e. $d < 2$, which is what damps the *large- $|z|$* region. Feynman-parametrise, shift $z \rightarrow z' + xy + \bar{x}y'$ with $\Delta = x\bar{x}r^2$, drop the odd term and use $z'^m z'^{m'} \rightarrow \frac{\delta^{mm'}}{d} z'^2$:

$$T_\epsilon^{mm'} = \mu^{2\epsilon} \int_0^1 dx \int d^d z' \frac{\frac{\delta^{mm'}}{d} z'^2 - x\bar{x} r^m r^{m'}}{[z'^2 + \Delta]^2} = \mu^{2\epsilon} \int_0^1 dx \left[\frac{\delta^{mm'}}{2} \pi^{d/2} \Gamma(\epsilon) \Delta^{-\epsilon} - x\bar{x} r^m r^{m'} \pi^{d/2} \Gamma(1+\epsilon) \Delta^{-1-\epsilon} \right].$$

Both x integrals give the same Euler beta $\int_0^1 (x\bar{x})^{-\epsilon} dx = \frac{\Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)}$, so

$$T_\epsilon^{mm'} = C(\epsilon) (r^2 \mu^2)^{-\epsilon} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right], \quad C(\epsilon) = \pi^{1-\epsilon} \frac{\Gamma(1+\epsilon) \Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)}.$$

Cross-check on the trace. Contracting with $\delta^{mm} = d = 2 - 2\epsilon$ gives $C(\epsilon) (r^2 \mu^2)^{-\epsilon} \frac{1-2\epsilon}{\epsilon}$, which is identical (symbolically, difference = 0) to the independent evaluation through the massless bubble $\int \frac{d^d z}{(z-y)^2 (z-y')^2}$ with $u \cdot v = \frac{1}{2}(u^2 + v^2 - r^2)$ and the two scaleless pieces set to zero. ✓

Expanding, $C(\epsilon) = \pi e^{-(\log \pi + \gamma_E - 2)}$, so with $\bar{\mu}^2 \equiv \pi e^{\gamma_E} \mu^2$

$$\frac{C(\epsilon) (r^2 \mu^2)^{-\epsilon}}{2\epsilon} = \frac{\pi}{2} \left[\frac{1}{\epsilon} - \log(r^2 \bar{\mu}^2) + 2 \right],$$

to be compared with $\frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right)$ from §3: the $-\log r^2$ matches, $-\pi \hat{r} \hat{r}$ matches, and

$$\frac{1}{\epsilon} \longleftrightarrow \log(R^2 \bar{\mu}^2) - 1.$$

This is the answer to your question. The pole and the $y \rightarrow y'$ logarithm are one and the same object, $\frac{1}{\epsilon} (r^2)^{-\epsilon}$. Dim reg does not "cure a UV divergence at $y \rightarrow y'$ " — there isn't one — it **resums the $\log(y-y')^2$ into a power and exposes the divergence as an infrared $\frac{1}{\epsilon}$ pole of the z integral**. Note the sign convention: $\epsilon > 0$ here, opposite to a UV subtraction.

5. Contracting $T_\epsilon^{mm'}$ with the momentum bracket

Write $X^k = \frac{(x-y)^k}{(x-y)^2}$, $X'^{k'} = \frac{(x'-y')^{k'}}{(x'-y')^2}$ and

$$\mathcal{X} \equiv X \cdot X' = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}, \quad \mathcal{R} \equiv (\hat{r} \cdot X)(\hat{r} \cdot X') = \frac{[r \cdot (x-y)][r \cdot (x'-y')]}{r^2 (x-y)^2 (x'-y')^2}.$$

The $2/k^+$ term dies. $T_\epsilon^{mm'}$ is symmetric (§3, §4), so $\delta_{mk'} \delta_{m'k}$ and $\delta_{mk} \delta_{m'k'}$ give the same contraction $T^{kk'} X^k X'^{k'}$ and

$$\frac{2}{k^+} (\delta_{mk'} \delta_{m'k} - \delta_{m'k'} \delta_{km}) T_\epsilon^{mm'} X^k X'^{k'} = 0$$

identically — confirmed numerically to 8×10^{-3} against a $\mathcal{O}(24)$ integrand (quadrature noise).

For the other two, with $d_\perp = 2 - 2\epsilon$ used consistently (it came from δ_{ii}):

$$\begin{aligned}\frac{d_{\perp} p^+}{S^2} \delta_{km} \delta_{k'm'} &\rightarrow \frac{p^+}{S^2} C(\epsilon) (r^2 \mu^2)^{-\epsilon} \left[\left(\frac{1}{\epsilon} - 1 \right) \mathcal{X} - 2\mathcal{R} \right], \\ \left(\frac{p^+}{k^+} + \frac{1}{p^+} \right) \delta_{mm'} \delta_{kk'} &\rightarrow \left(\frac{p^+}{k^+} + \frac{1}{p^+} \right) C(\epsilon) (r^2 \mu^2)^{-\epsilon} \left(\frac{1}{\epsilon} - 2 \right) \mathcal{X}.\end{aligned}$$

Adding them, with $\bar{L} \equiv \frac{1}{\epsilon} - \log(r^2 \bar{\mu}^2)$, the scheme constants $+2$ and -2 **cancel** and everything collapses to

$$\pi \left\{ \frac{\bar{L} \mathcal{X}}{p^+} + \frac{p^+}{k^+} \bar{L} \mathcal{X} + \frac{p^+}{(k^+ + p^+)^2} [(\bar{L} + 1) \mathcal{X} - 2\mathcal{R}] \right\}$$

(symbolic difference from the direct ϵ -expansion: exactly 0). The residual $\frac{1}{\epsilon}$ pole therefore has coefficient $\mathcal{X} \left[\frac{1}{p^+} + \frac{p^+}{k^+} + \frac{p^+}{(k^+ + p^+)^2} \right]$ and multiplies **exactly the leading-order transverse structure** $\mathcal{X} [U(y') - U(x')][U(y) - U(x)]$.

6. The p^+ integral, and the $+$ prescription

All the remaining p^+ dependence is the bracket above times the phase $e^{-i\kappa p^+}$, $\kappa = \frac{k \cdot (y' - y)}{k^+}$. Three integrals, $P_{\vee} = \vee - k^+$:

6.1 The $\frac{1}{p^+}$ term — here the $+$ prescription is legitimate

Because the z integral was **already made finite by dim reg**, the coefficient of $\frac{1}{p^+}$ is a p^+ -independent, transversally finite number; and the $p^+ \rightarrow 0$ limit of the phase does **not** strip the leading-order oscillation, since $e^{-i\kappa \cdot (y' - y)}$ factored out earlier. That is precisely the condition that failed in the earlier rows. So, exactly (up to $O(\kappa \Lambda)$):

$$\mathcal{I}_1 = \int_{\Lambda}^{P_{\vee}} \frac{dp^+}{p^+} e^{-i\kappa p^+} = \underbrace{\log \frac{\vee - k^+}{\Lambda}}_{\text{rapidity log}} + \underbrace{\int_0^{\vee - k^+} \frac{dp^+}{p^+} [e^{-i\kappa p^+} - 1]}_{[1/p^+]_+},$$

$$\mathcal{I}_1 = \log \frac{\vee - k^+}{\Lambda} - \text{Cin}(|\kappa| P_{\vee}) - i \text{sgn}(\kappa) \text{Si}(|\kappa| P_{\vee}), \quad \text{Cin}(x) = \gamma_E + \log x - \text{Ci}(x).$$

kappa=0.70	Pv=20.0	Lam=1e-06	quad=(13.664366 - 1.5562104j)	closed=(13.664366 - 1.5562111j)
kappa=2.30	Pv=50.0	Lam=1e-07	quad=(14.716215 - 1.5735573j)	closed=(14.716215 - 1.5735576j)

6.2 The $\frac{p^+}{k^+}$ term — regular at $p^+ = 0$, no prescription needed

$$\mathcal{I}_2 = \int_0^{P_{\vee}} dp^+ p^+ e^{-i\kappa p^+} = \frac{1 - e^{-i\kappa P_{\vee}} (1 + i\kappa P_{\vee})}{-\kappa^2}.$$

6.3 The $\frac{p^+}{(k^+ + p^+)^2}$ term

Substituting $q = k^+ + p^+$ (so $q : k^+ \rightarrow \vee$):

$$\mathcal{I}_3 = \int_0^{P_{\vee}} \frac{p^+ e^{-i\kappa p^+}}{(k^+ + p^+)^2} dp^+ = e^{i\kappa k^+} \left[(1 + i\kappa k^+) (\text{Ei}(-i\kappa \vee) - \text{Ei}(-i\kappa k^+)) + k^+ \left(\frac{e^{-i\kappa \vee}}{\vee} - \frac{e^{-i\kappa k^+}}{k^+} \right) \right].$$

kappa=0.70	k+=1.0	V=20.0	quad=(0.059512843 - 0.34483599j)	closed=identical	diff=3.1e-16
kappa=2.30	k+=0.4	V=50.0	quad=(-0.018779326 - 0.29950478j)	closed=identical	diff=2.8e-16
kappa=0.13	k+=2.0	V= 9.0	quad=(0.633395130 - 0.30927117j)	closed=identical	diff=1.1e-16

7. The result

$$\begin{aligned}\frac{d^3 N^{\text{NLO}}}{d^3 k} \Big|_{\text{cc}} &= \frac{1}{(2\pi)^3} \frac{g^4 N_c}{4\pi^4} \frac{1}{2k^+} \int_{x, x', y, y'} e^{-i\kappa \cdot (y' - y)} \left\{ \bar{L} \mathcal{X} \left[\mathcal{I}_1 + \frac{\mathcal{I}_2}{k^+} \right] + [(\bar{L} + 1) \mathcal{X} - 2\mathcal{R}] \mathcal{I}_3 \right\} \\ &\quad \times [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x)\end{aligned}$$

with $\bar{L} = \frac{1}{\epsilon} - \log(r^2 \bar{\mu}^2)$, $\bar{\mu}^2 = \pi e^{\gamma_E} \mu^2$, $r = y' - y$, $\kappa = \frac{k \cdot r}{k^+}$, and $\mathcal{X}, \mathcal{R}, \mathcal{I}_{1,2,3}$ as defined in §5–6. The $\frac{\pi}{2\pi} = \frac{1}{2}$ is the $C(0) = \pi$ of the z integral against the $\frac{dp^+}{2\pi}$.

7.1 The rapidity logarithm

Only $\mathcal{I}_1$ carries $\log \frac{\sqrt{-k^+}}{\Lambda}$, and it rides on $\bar{L} \mathcal{X} [U(y') - U(x')][U(y) - U(x)]$ — the leading-order structure. Writing $\log \frac{\sqrt{-k^+}}{\Lambda} = \log \frac{\sqrt{-}}{\Lambda} + \log \left(1 - \frac{k^+}{\sqrt{-}}\right)$ the first piece is absorbed into JIMWLK exactly as in the previous rows; the second is a finite $\sqrt{-}$ -dependent remainder.

7.2 The $\frac{1}{\epsilon}$ is infrared — track it

The pole is **not** renormalised away by a coupling or a DGLAP splitting: it came from $|z| \rightarrow \infty$ and it sits on the LO colour structure. It exists in this row alone only because $U^{db}(z)U^{db}(z) = \delta^{dd}$ wiped out every $U(z)$. In rows where a $U(z)$ survives, $U(z) \rightarrow 1$ at large $|z|$ cuts the same integral off. So the $\frac{1}{\epsilon}$ here must cancel against the companion rows in the full NLO sum — worth checking explicitly before using this row on its own. Note also the mixed double logarithm $\frac{1}{\epsilon} \log \frac{\sqrt{-k^+}}{\Lambda}$ hiding in $\bar{L} \mathcal{I}_1$.

7.3 If you prefer the cutoff form

Replace $\bar{L} \rightarrow \log \frac{R^2}{r^2} + 1$ and $(\bar{L} + 1) \rightarrow \log \frac{R^2}{r^2} + 2$; every other line is unchanged.

zint/src/row8_algebra.py (§1), row8_colour.py (§1.4), row8_transverse.py (§2–3), row8_dimreg.py (§4–5), row8_final.py (§5–6) — every number above.